Trajectory-Aware Reinforcement Learning for Autonomous Tractor-Trailer Reverse Docking

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Project Goal

Reversing a tractor-trailer into a loading dock is a difficult control problem because the vehicle is nonlinear, nonholonomic, underactuated, and unstable when moving backward. Small steering errors make the articulation angle grow, which can lead to a ‘jackknife’, when the trailer swings around until it folds against the tractor at a sharp angle. Our goal is to build a simulated system that plans and executes a safe reverse-docking maneuver at a logistics yard. Given the tractor and trailer state, a static map, and the pose of a selected loading door, the system must place the trailer within a position and heading tolerance while avoiding obstacles, respecting steering limits, and keeping the articulation angle within a safe bound. Our research question is whether a Proximal Policy Optimization (PPO) controller that uses trajectory-relative preview information can reliably reverse a tractor-trailer into a dock, and how it compares with a model-based controller when the starting pose, trailer geometry, and observation noise change. We expect trajectory-relative observations to be easier to learn from than raw global coordinates, the model-based controller to be strong in nominal conditions, and PPO to offer faster online control and more tolerance to moderate model mismatch.

Simulation Platform

We will first develop the task in a lightweight 2D environment that follows the Gymnasium API, so the vehicle model, reward, and safety checks can be tested quickly. We will then transfer both controllers to a 3D MuJoCo simulation containing an articulated tractor with a hinged semitrailer and a logistics-yard scene with loading doors and static obstacles. Simulator ground truth will provide localization so that planning and control can be evaluated on their own.

ML/RL Method

Our main method is Proximal Policy Optimization (PPO), an actor-critic policy gradient algorithm. PPO fits this task because the action is a continuous steering command, which rules out tabular Q-learning and DQN without coarse discretization. Its clipped update is stable and not very sensitive to hyperparameters, which matters for an unstable system where one bad update can make the policy jackknife on every episode. It also works well with many parallel randomized episodes, and simulation is cheap here, so PPO's lower sample efficiency compared with off-policy methods like SAC is not a major drawback. For comparison we will implement a nonlinear model predictive controller (NMPC) that minimizes tracking, articulation, and steering costs subject to the vehicle constraints.

Programming Language and Tools

We will use Python 3 with Gymnasium for the 2D environment, MuJoCo and its Python bindings for the 3D simulation, Stable-Baselines3 and PyTorch for PPO, CasADi with the IPOPT solver

for NMPC, NumPy and SciPy for the vehicle model and geometry, and Matplotlib and TensorBoard for plots and training logs. Code will be shared through GitHub.

Project Description

The robot is one tractor towing one conventional semitrailer at low reverse speed. The vehicle is modeled with a kinematic single-track model with an on-axle hitch. The state includes tractor position and heading, trailer heading, articulation angle, steering angle, and velocity. An episode ends when the trailer reaches the dock within tolerance, when any part of the vehicle collides with an obstacle, when the articulation angle exceeds its limit (a jackknife), or at a time limit. A planner first supplies predefined straight and curved reference paths to the dock, and later a Reeds-Shepp or Hybrid A* generator if time permits. Since a car-like path does not guarantee a safe articulation history for the trailer, every generated path is checked by simulating the tractor-trailer model along it. Figure 1 shows how the parts connect.

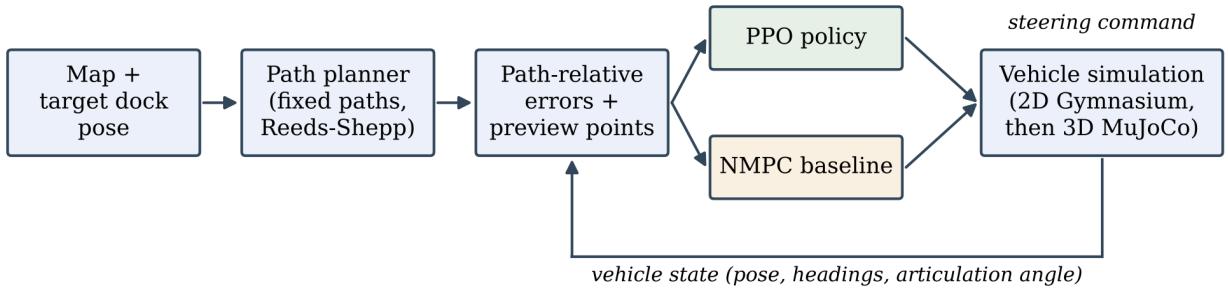


Figure 1. System overview. The planner produces a reference path, path-relative errors and preview points are computed from the vehicle state, and either the PPO policy or the NMPC baseline sends steering commands to the same 2D or MuJoCo vehicle through a shared interface.

The PPO observation is measured relative to the reference path rather than in global coordinates. It contains the trailer lateral error, tractor and trailer heading errors, articulation angle, current steering angle, distance to the goal, and the lateral and heading offsets of several preview points ahead on the path. The action is a bounded steering-rate command at a fixed low reverse speed, which keeps steering smooth. The reward combines progress along the path, a penalty on tracking error and articulation angle, a small penalty on steering rate, a large bonus for a successful dock, and large penalties for collision or jackknife. Safety limits are also enforced by episode termination and a simple supervisor that clips unsafe commands, so safety does not depend only on reward shaping. Training uses domain randomization over the starting pose, trailer length, and sensor noise.

A purely classical controller such as NMPC can work very well, but it needs an accurate model and solves an optimization problem at every step. End-to-end RL from global coordinates must relearn the geometry for every dock location and tends to train slowly. Imitation learning would need a large set of expert demonstrations that we do not have. Separating planning from control

and giving the policy trajectory-relative inputs makes the learning problem smaller and independent of where the dock is, while still keeping a fast learned controller. We will test this claim directly by training one PPO policy with global-coordinate observations and one with trajectory-relative observations.

Both controllers will be run on at least 100 held-out trials per condition with identical starting states and target docks. Metrics are docking success rate, collision rate, jackknife rate, final lateral, longitudinal, and heading error, maximum articulation angle, maneuver time, path length, steering smoothness, and controller computation time. We will compare PPO with NMPC, global with trajectory-relative observations, and perfect with noisy state information, and we will vary the starting pose and trailer geometry. Failed trials will be reported and categorized, not dropped. Our minimum target is at least 90% success in nominal held-out scenarios, no collisions in successful trials, no articulation-limit violations, and a reproducible full docking maneuver in MuJoCo.

Fallback Plan

Should the automatic planner not be working early in the project, we will use only the fixed straight and curved reference paths. For the baseline, a simpler pure-pursuit or LQR tracking controller will replace NMPC if it does not solve reliably in real time. In case PPO cannot reach full docking, we will train it with a curriculum that starts from short straight reverses near the dock and report results on the subset of maneuvers it does solve. A delayed MuJoCo transfer would mean running the full quantitative comparison in 2D and using MuJoCo to demonstrate a smaller set of scenarios. Each of these options still produces a complete PPO versus model-based comparison on reverse trailer tracking before the end of the course. Simulated LiDAR mapping and camera-based dock-number recognition are optional extensions and will be dropped first.

References

Yan et al. (2024) is the closest work and shows that trajectory-based state representations help RL learn truck-trailer reverse driving. We build on this idea but add a dock target, obstacles, a 3D MuJoCo transfer, and a direct comparison against NMPC. Moradi and Mehrandezh (2022) present an NMPC local planner for tractor-trailers in forward and backward motion, which is the basis for our baseline. Ma et al. (2025) combine graph search with RL for tractor-trailer planning; our work uses a similar split between planner and learned controller but focuses on the low-level tracking controller and its robustness. Menz and Wenzel (2026) use imitation learning for tractor-trailer positioning, which gives a contrasting learning approach that needs demonstrations, while our method learns from reward alone.

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